Math SL Internal Assessment

Related Rates in a Soccer Game

I. Introduction

When I was in kindergarten, my teacher told my parents that I was uncoordinated, so they should sign me up for a sport. Concerned, my parents immediately signed me up for a soccer team, and I have been playing ever since. Soccer has been a major part of life, so when I learned related rates in calculus, I was excited because I felt like I had gained knowledge about something that I could apply to my own life, specifically soccer. I play an offensive position (forward and center midfield), and my position on the field allows me to have various scoring opportunities. Goals can arise from shots, corner kicks, breakaways, and through balls.

Aim: I will investigate the related rates that could be found in a shot, corner kick, breakaway, and through ball.

In order for this investigation to be even more applicable and realistic, data was collected from my varsity high school goalkeeper, and a varsity high school offensive player. The data is displayed below in Table 1, 2, and 3 and will be used throughout the investigation.

Table 1: Data Collected from High School Varsity Goalkeeper

This data will be used throughout the investigation, specifically in the related rates with the goalkeeper. Each related rate will first be solved in general, then the equation will be evaluated with these values when it is applicable.

	Goalkeeper
Height (yards)	1.89
Height with hands ready to catch (yards)	0.8472

Table 2: Data Collected From Both High School Varsity Goalkeeper and Offensive Player

A goalkeeper and offensive player from the high school varsity team was asked to participate in this investigation. They completed a series of eighteen-yard running exercises, so that the data could be used when evaluating the related rates. Both the player and keeper ran three eighteen yard sprints, three sprints with the ball, three shuffling with their left foot forward, and three shuffling with their right foot forward. The times for each of these eighteen-yard sprints were used to calculate the players' speed for each running exercise, and the average of these speeds will be used when evaluating the related rates. The eighteen-yard sprint was used to calculate the player's top sprinting speed. The eighteen-yard sprint with the ball was used to calculate their speed when dribbling the ball, since when players dribble, they tend to go slower. The eighteen-yard shuffle, with both the left and right foot was used to calculate the shuffling speed of the player, and will be used in the corner related rate.

	Goalkeeper time (sec)	Goalkeeper speed (yd/sec)		Offensive Player time (sec)	Offensive Player speed (yd/sec)
18 yd sprint	3.25	5.54		2.88	6.25
	3.13	5.75		3.11	5.79
	3.27	5.50		3.03	5.94
average:	3.22	5.60		3.01	5.99
18 yd sprint with the ball	3.80	4.74		3.50	5.14
	3.68	4.89		3.28	5.49
	3.48	5.17		3.70	4.86
average	3.65	4.93		3.49	5.15
18 yd shuffle (leading with left foot)	5.01	3.59		4.78	3.77
	4.57	3.94		4.74	3.80
	4.94	3.64		4.91	3.67
average	4.84	3.72		4.81	3.74
18 yd shuffle (leading with right foot)	4.70	3.83		4.78	3.77
	4.68	3.85		4.81	3.74
	3.97	4.53		4.54	3.96
average	4.45	4.04		4.71	3.82

Table 3: Data Collected From High School Soccer Goal

The dimensions of a soccer goal were first measured in inches, and then converted to yards in the table below. This collected data will be used in the shot related rate since the goal dimensions are needed to evaluate this particular related rate.

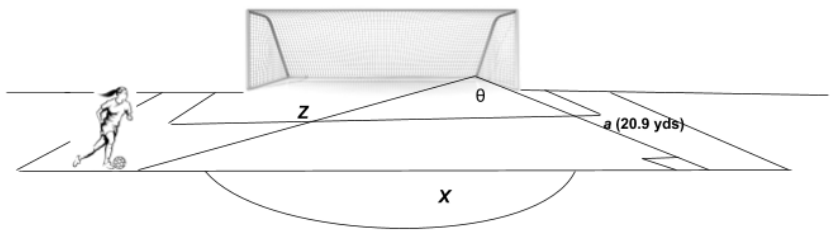
dimensions of goal	length	height	depth
inches	289.75	99.89	104.32
yards	8.05	2.77	2.90

Part II: The Shot

When the ball is in the offensive third (which is about thirty yards from the end line), the attacking team is going to take as many shots as possible. The intensity of play increases as the ball moves closer to the goal, and this aggressive attacking makes the game really exciting. A very common play that emerges is passing the ball to a forward, who then turns, dribbles and shoots to the far post. However, players will sometimes continue to dribble, and coaches say this diminishes their chances for a shot. For example, a player who receives the ball at the edge of the eighteen-yard box (see Figure A) may continue to dribble to the right. Hence the question, after 1

second, at what rate is the angle (θ) changing? At what rate is the angle changing after 2 seconds? Figure A illustrates the situation.

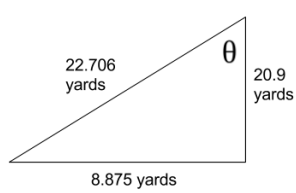
Figure A:
a = constant (distance from back corner of the goal; this was determined using the goal dimensions in Table 3)
*When the player receives the ball, *X* is 14.025 yards in length. This was determined using the goal dimensions in Table 3, and the dimensions of a soccer field.[1]



Using the figure above, and *a* as a constant, $\frac{d\theta}{dt}$ was found in general with the work below:

$$\tan\theta = \frac{X}{a}$$
$$\frac{d\theta}{dt}(\tan\theta) = \frac{da}{dt}\left(\frac{1}{a}\frac{dX}{dt}\right)$$
$$\frac{d\theta}{dt}(\sec\theta)^2 = \frac{1}{a}\frac{dX}{dt}$$
$$\frac{d\theta}{dt} = \frac{1}{a}\frac{dX}{dt}(\cos\theta)^2$$

To evaluate this equation after 1 second, the data from Table 2 and 3 was used. Since $\frac{dX}{dt}$ is the speed of the offensive player when he or she is dribbling the ball, -5.15 yards/sec was plugged in for $\frac{dX}{dt}$ which is the speed of the specific high school offensive player (from Table 2). *A* is 20.9 yards because this is the constant (this was determined using the goal dimensions in Table 3 and the standard dimensions of a soccer field). Using the speed of the offensive player with the ball, the triangle above was evaluated at this specific moment in time: when *t* = 1 sec. The triangle below represents this moment in time.



Therefore, $(\cos\theta)^2$ was found to be 0.847, and this was plugged into the equation.

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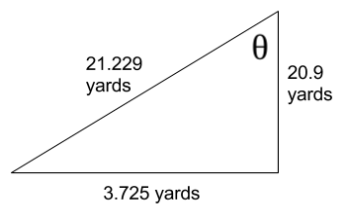
[1] Josip Punis, "What are the official dimensions of a soccer field in the FIFA World Cup?," What

$$\frac{d\theta}{dt} = \frac{1}{20.9} \times (-5.15) \times (0.847) = -0.209\, rad/sec$$

Therefore, at 1 second, the angle for the shot is decreasing at a rate of -0.209 rad/sec.

To evaluate this equation after 2 seconds, the data from Table 2 and 3 was used. Since $\frac{dX}{dt}$ is the speed of the offensive player when he or she is dribbling the ball, -5.15 yards/sec was plugged in for $\frac{dX}{dt}$. -5.15 yards/sec is the speed of the specific high school varsity offensive player (see Table 2). A is 20.9 yards because this is the constant (this was determined using the goal dimensions in Table 3 and the standard dimensions of a soccer field). Using the speed of the offensive player with the ball, the triangle above was evaluated at this specific moment in time: when t = 2 sec. The triangle below represents this moment in time.



Therefore, $(\cos\theta\ )^2$ was found to be 0.970, and this was plugged into the equation.

$$\frac{d\theta}{dt} = \frac{1}{20.9} \times (-5.15) \times (0.969) = -0.238\, rad/sec$$

Therefore, at 2 seconds, the angle for the shot is decreasing at a rate of -0.238 rad/sec.

Limitations

In this problem there are several assumptions and simplifications made. For example, the average speed was used, so this doesn't take into account any change in speed on the offensive player's part. Additionally, the offensive player is assumed to have dribbled in a straight line. In a real game, a forward may have to beat players, and this would have changed his or her speed and path as well. This problem also does not take into account any movement of the goalkeeper, or any special curve or spin a player can put on a ball; soccer players can curve the ball so their angle of shot is not so limited.

Reflection

Nevertheless, this investigation was worthwhile. Deriving the general equation allows various speeds to be plugged in for different players. Additionally, coaches advise players to take a shot early and fast when they are in the eighteen-yard box, and not to dribble away or parallel to the goal because it diminishes the angle of the shot. This problem reveals the math behind their advice. As a player continues to dribble parallel to the goal, and more time passes, the magnitude of $\frac{d\theta}{dt}$ increases, as seen in the comparison of $\frac{d\theta}{dt}$ evaluated after one second, and then evaluated after two seconds. The angle for a shot to the far post is diminishing faster as the player continues to hold on to the ball. This makes the shot much harder to take, and the mechanics behind the shot increase in difficulty as well. The player will have to turn his or her hips more in order to get the shot off. In addition, not only does this give the keeper more time to get into a more advantageous position, but it also makes the save easier. The angle of the shot is smaller, so the goalkeeper can better anticipate where the ball is going to go as the player continues to dribble. As a result, the optimal shot arises when the player is farther away from the corner he or she is aiming for. If a player receives a ball on one side of the eighteen-yard box, they are told to shoot

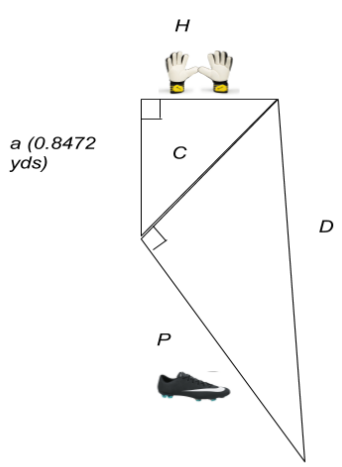
the ball towards the far post. As supported by the related rate, as a player continues to dribble, the angle theta decreases, so theta is greatest when the player receives the ball.

This investigation into the shot revealed how imperative it is for the player to take the shot once they receive the ball. The angle for the shot is decreasing when they begin to dribble, and at a fairly rapid rate, even after only one second. Because the related rate was evaluated again after two seconds, it was revealed that the angle for the shot is decreasing even faster. It is important to know these values because they quantify coach's advice to not keep dribbling the ball. The related rate also gives a value to this specific offensive player. Because the related rate was solved in general first, any player can input his or her sprinting speed with the ball for $\frac{dX}{dt}$, or different values for a as well, so they can determine for themselves how fast the angle for the shot is changing. These values can be used to improve a player's shot. If they can understand at what time the angle for the shot is decreasing too quickly, then the player can know that they must shoot before this time.

Part III. The Corner Kick

Other than a shot, one of the most dangerous plays in a soccer game is the corner kick. With a corner kick, the attacking team should have a very decent chance of finishing the goal. Because a corner kick is so valuable, coaches will devise plays that include the runs by the offensive players. Many coaches will place a player in front of the keeper to block his or her sight of the ball when the corner is kicked. This led to the questioning of the relationship between the keeper and the player. In particular, the relationship between the keeper's hands and the player's feet – the two body parts that could determine the outcome of the game. Therefore, the question arose, if the offensive player and keeper move simultaneously once the corner is kicked, how fast will the player's feet and keeper's hands be separating 1.1 seconds later? Figure B illustrates the situation below.

Figure B
H = hands of the keeper
a = height of keeper's hands off the ground (constant; from the data in Table 1)
P = player's feet



$$H^2 + a^2 = C^2$$
$$C^2 + P^2 = D^2$$

Using substitution for C^2:

$$H^2 + a^2 + P^2 = D^2$$

Take the derivative:

$$\frac{d}{dt}(H^2 + a^2 + P^2) = \frac{d}{dt}(D^2)$$
$$2H\frac{dH}{dt} + 0 + 2P\frac{dP}{dt} = 2D\frac{dD}{dt}$$

Therefore, in general

$$\frac{H\frac{dH}{dt}+0+P\frac{dP}{dt}}{D} = \frac{dD}{dt}$$

To evaluate the above equation after 1.1 seconds, the data from Table 1 and 2 was used. After consulting with the varsity goalkeeper and offensive player, it was determined that the players would shuffle in this situation. Since the kick is coming from the right, in this situation, the keeper's shuffle speed leading with their left foot, 3.72 yards/second (from Table 2), was plugged in for $\frac{dH}{dt}$. The offensive player's shuffle leading with their right foot was used because they are moving away from the keeper. So, 3.82 yards/second (from Table 2) was plugged in for $\frac{dP}{dt}$. These rates were used to determine *H, D,* and *P* at this particular moment (after 1.1 seconds). Thus, 4.092 yards was plugged in for *H,* 4.202 yards was plugged in for *P,* and 5.926 yards was plugged in for *D.*

$$\frac{(4.092\times3.72) + (4.202\times3.82)}{5.926} = \frac{dD}{dt} = 5.28\ yards/second$$

Therefore, after 1.1 seconds, the distance between the goalkeeper's hands and the player's foot is increasing at a rate of 5.28 yards/second.

Limitations

Like Part II, the shot, there are several assumptions made. For example, the hands of the goalkeeper are not directly above the player's feet before they separate, however, there could most definitely be contact between the keeper and the player that would bring the hands of the keeper almost directly above the player's feet. In addition, it assumes a linear, and perpendicular movement by both the keeper and the player, which may not happen every single time in a game.

Reflection

Nonetheless, the equation demonstrates the relationship between a keeper's hands, and the player's foot. Also, deriving the general equation allows various speeds to be plugged in for different players. This relationship really highlights how important size can be in the game. For example, if the keeper was taller, or held his or her hands up farther, $\frac{dD}{dt}$ would be smaller since the keeper's height is used to determine *D,* which is in the denominator. Therefore, a taller keeper/one who holds their hands higher would result in a slower rate. This height relationship then questions the best mode of protecting the goal. Is it better to be lower to the ground, so the keeper can react to low shots faster? Perhaps, it is better to be taller so the distance between the initial shooting point (the foot) and the keeper's hands is not separating so quickly. With keepers tending to be larger, because they obviously take up more space in the goal, this related rate really supports the positives in having a tall and big goalkeeper.

Part IV. The Breakaway

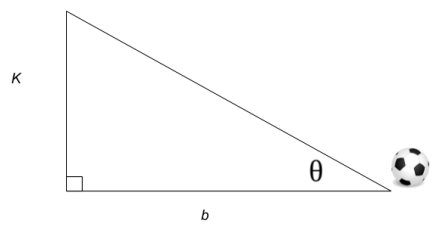
The breakaway is another chance for an offensive player to score a goal, and can be the most exciting. In a breakaway, the offensive player is in a one versus one situation, in which only the keeper, sometimes, another defender, stands between the offensive player and the goal.

A successful breakaway is a testimony to the skill and speed of an offensive player, and a failed breakaway will repeatedly play on the sports highlights. The offensive player approaching the keeper has to decide whether or not he or she should shoot or dribble the ball. An effective shot is the chip shot, in which the offensive player kicks the ball over the keeper, and into the goal. This chip shot breakaway situation led to the question: if a forward, 20 yards from the keeper, is dribbling towards a keeper, after 2 seconds, what is the rate of change of the angle for the shot? After 2.5 seconds, what is the rate of change of the angle for the shot?

Figure C

b = dribbled ball; initially, the length of *b* is 20 yards

K= keeper height (constant); refer to table 1



$$\tan\theta = \frac{K}{b}$$

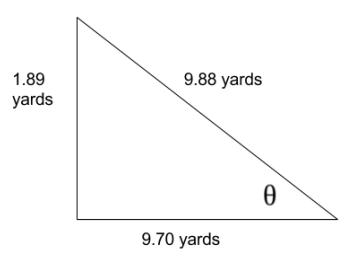
$$\frac{d}{dt}(\tan\theta) = \frac{d}{dt}\left(\frac{K}{b}\right)$$

$$(\sec\theta)^2\frac{d\theta}{dt} = K\frac{1}{b}\frac{db}{dt}$$

Therefore, in general, the change in the angle of the shot is represented by the function below:

$$\frac{d\theta}{dt} = \frac{K\frac{db}{dt}\cos\theta^{\,2}}{b^2}$$

To evaluate the above equation after 2 seconds, the data from Table 1 and 2 was used. Because the offensive player is sprinting with the ball, the average sprinting speed with the ball was used, so 5.15 yards/second (from Table 2) was plugged in for $\frac{db}{dt}$. 1.89 yards (from Table 1) was plugged in for *K,* because that is the keeper's height. $(\cos\theta)^2$ was evaluated at that moment, *t*=2 seconds, which is illustrated in the triangle below.



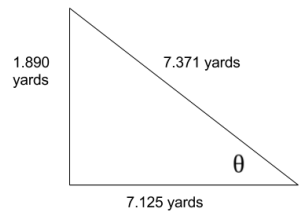
Therefore, 0.963 was plugged in for $(\cos\theta)^2$. Using $\frac{db}{dt}$, *b* was evaluated at 2 seconds as well, so 9.70 yards was plugged in for *b*.

$$\frac{d\theta}{dt} = \frac{(1.89)\times(5.15)\times(0.963)}{(9.70)^2} = 0.100\ rad/sec$$

Therefore, after 2 seconds, the angle for a chip shot is changing at a rate of 0.100 rad/sec.

To evaluate the above equation after 2.5 seconds, the data from Table 1 and 2 was used. Because the offensive player is sprinting with the ball, the average sprinting speed with the ball was used, so 5.15 yards/second (from Table 2) was plugged in for $\frac{db}{dt}$. 1.89 yards (from Table 1) was

plugged in for *K,* because that is the keeper's height. $(cos\ \theta\ )^2$ was evaluated at that moment, *t*=2.5 seconds, which is illustrated in the triangle below.



Therefore, 0.934 was plugged in for $(cos\ \theta\ )^2$. *B* was evaluated at 2.5 seconds as well, so 7.125 yards was plugged in for *b.*

$$\frac{d\theta}{dt} = \frac{(1.89)\times(3.49)\times(0.973)}{(7.125)^2} = 0.179\ rad/sec$$

Therefore, after 2.5 seconds, the angle for a chip shot is changing at a rate of 0.179 rad/sec.

Limitations

There were two major assumptions made, and they were that the keeper would stay where they are, and the ball would travel in a straight line when chipped. Most keepers will run out to meet the ball as the offensive player is dribbling towards them, and a chip follows a curved trajectory.

Reflection

Despite this, deriving the general equation allows various speeds and distances to be plugged in for different players and plays. Additionally, the $\frac{d\theta}{dt}$ equation is still valuable to investigate, and illustrates the mechanics and difficulty behind the chip shot. If the equation $\tan\theta\ = \frac{K}{b}$ were to be evaluated at 2 seconds and at 2.5 seconds, it is found that the angle is greater at 2.5 seconds, which makes sense since the ball is closer to the keeper. The interesting thing however, is the rate at which the angle is changing. The angle is changing faster at 2.5 seconds in comparison to 2 seconds, demonstrating how with one big touch, or shooting the ball too late, the chip shot will fail since the angle of the shot is too steep. Moreover, because of this, as supported by the related rate, the timing of the chip shot is imperative. Too soon, and the keeper can back up and catch the ball, too late, and the shot is going to hit the keeper. However, used correctly, the chip shot can be extremely effective. As demonstrated through the related rate, even half a second can determine whether or not the play is successful for the offensive player. In the game, a player is obviously not calculating the related rate, however, an offensive player does his or her own intuitive calculations in order to determine when to shoot, dribble, or pass, in this situation.

This investigation is important because it quantifies, for this specific offensive player, how their angle for chip shot is changing. The related rate puts a value on how fast the angle for the shot is increasing as they approach the keeper. For the offensive player, this is valuable knowledge. By knowing the rate at which their angle of shot is changing, they will be able to determine whether or not they should take the chip shot. They can use this related rate to determine at what time the angle for the chip shot is changing too fast, making the chip shot impossible. Therefore, they can determine that after *t* seconds, the offensive player must chip the ball at this moment, because if they hold on to the ball any longer, the angle for the chip shot will be increasing too quickly, and the chip shot will fail. The offensive player can then use this to determine at what time they should dribble around the keeper, shoot around the keeper, or even

pass the ball.

Part V. The Through Ball

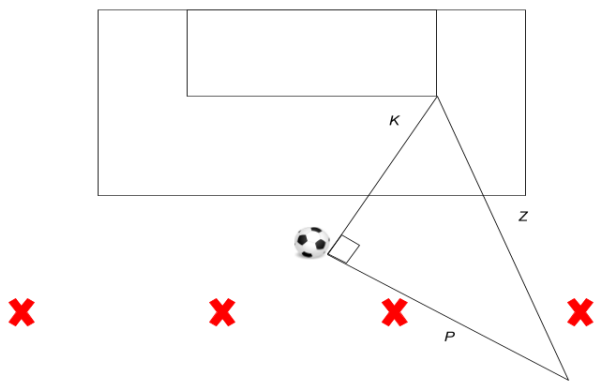
One other very good offensive chance for an offensive player is the through ball. A through ball is a pass threaded through the defense of an opposing team to another player on the run. For example, center backs can be split, in which the ball is passed in between them, and another player, on the opposite team, is able to run and get the ball. Through balls are very effective, and can lead to breakaways, leading to a race between players as each sprints to get the ball. The most effective through balls can set a player up to a foot race between the keeper and the offensive player. This situation presents a relationship between the distance between the keeper and player, and leads to the question: after 2 seconds, at what rate is the distance between the keeper and the player changing? Figure D illustrates the described situation below.

Figure D

Z = distance between the keeper and player

X = defensive players

* When time is zero, *P* is 20 yards long, and *K* is 17 yards long.



$$Z^2 = P^2 + K^2$$

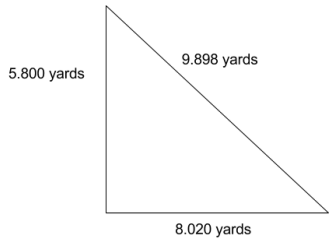
After taking the derivative,

$$2Z\frac{dZ}{dt} = 2P\frac{dP}{dt} + 2K\frac{dK}{dt}$$

$$Z\frac{dZ}{dt} = P\frac{dP}{dt} + K\frac{dK}{dt}$$

$$\frac{dZ}{dt} = \frac{P\frac{dP}{dt} + K\frac{dK}{dt}}{Z}$$

To evaluate the above equation after 2 seconds, the data from Table 2 was used. Because the player and the keeper were both sprinting towards the ball, their average sprinting times were used from Table 1. Thus, -5.99 yards/second was plugged in for $\frac{dP}{dt}$, because this is how fast the player is sprinting, and -5.60 yards/second was plugged in for $\frac{dK}{dt}$, because this is how fast the keeper is sprinting. Each of these rates were evaluated at the moment, *t*=2 seconds. The triangle below illustrates the moment in time.



Thus, 8.02 yards was plugged in for *P,* 5.8 yards was plugged in for *K,* and 9.898 yards was plugged in for *Z.*

$$\frac{dz}{dt} = \frac{(8.02)\times(-5.99) + (-5.80)\times(5.6)}{(9.898)} = -8.135\ yards/second$$

Therefore, after 2 seconds, the distance between the keeper and the offensive player is changing at a rate of -8.135 yards/second.

Limitations

The major limitation to this specific investigation is the initial distance between the player and the ball and the keeper and the ball are completely random. Additionally, the keeper is usually not standing at the corner of the six-yard box as Figure D portrays. The only time a keeper is usually in that position is after a goal kick, but other than that particular situation, the keeper is usually in the center of his or her goal. This also assumes that the ball rolls to a stop, but usually in a soccer game, the ball never stops. Moreover, this situation also does not take into account any movement of the defensive line. A center back will try to block the offensive player, and may get physical with the other player, which would not make the offensive player's route a straight line, and it would slow them down.

Reflection

This related rate really demonstrates how exciting soccer can be. The distance between the player and the keeper is decreasing at a very fast rate, and as the distance between the keeper and the player decreases, the tension on the field rises. Who will get to the ball first? Will there be a foul? This related rate embodies the excitement felt by the players and the fans as they wait with bated breath to see who will arrive to the ball, and potentially win the game. This related rate also illustrates how the through ball is very effective and dangerous, since it truly is a foot race to get to the ball. Fortunately, offensive players are usually faster than keepers.

Additionally, the general equation allows various speeds and distances to be plugged in for different players and plays. The through ball presents a chance for a race between a keeper and an offensive player. Normally, soccer players that play in offensive positions are extremely fast, so the through ball utilizes this speed in the attacking third. Analyzing this related rate also illuminates the fast decision making on both the keeper and player's side. If the keeper hesitates to come out for even one second, they could be beaten to the ball, and won't make the save. A player must also quickly decide if he or she wants to dribble and beat the keeper, attempt to chip the keeper, or try to shoot the ball around the goalkeeper. The offensive player must very quickly analyze the optimal angle for a shot, or time for a move around the keeper. In his or her own instinctual way, an offensive player is calculating the rate at which the distance between the keeper and the goal is changing. The longer a player delays, their play options also diminish. As the distance between the opposing players dwindles, the offensive player's options for a shot also decrease as well, and the keeper's opportunity for a save increases. Analyzing this change in distance happens very quickly for athletes, and the situation described above could happen anywhere on the field, not just in front of the goal. For example, center midfielders decide when to do a certain move depending on how fast the distance between them and the opposing player is

changing. Gauging the changing distance between an opposing player and themselves is important for all offensive players. As the distance between the offensive player and the opposing player diminishes, their options concerning what to do with the ball also diminish. Thus, knowing this related rate gives players the opportunity to use it to study and learn the most optimal times to perform certain moves and plays. For example, an opposing player needs more time, distance, and changing distance to complete a move in comparison to the distance, changing distance, and time they need to toe poke the ball. When the offensive player is sprinting towards the goal, they understand that they must get through, or around, the keeper in order to score. Therefore, by knowing this related rate, the offensive player is aware of the choices that they may have to beat the keeper. They know that as the distance between them and the keeper is decreasing at a rate of 8.135 yards/second, they will have time to, let's say, slide tackle the keeper. Of course, every player has a unique range of time and distance needed for moves.

V. Conclusion

In this investigation, it was found that for Part II (the shot), at 1 second, the angle for the shot is decreasing at a rate of -0.209 rad/sec; at 2 seconds, the angle for the shot is decreasing at a rate of -0.238 rad/sec. For Part III (the corner kick), after 1.1 seconds, the distance between the goalkeeper's hands and the player's foot is increasing at a rate of 5.28 yards/second. In Part IV (the breakaway), it was found that after 2 seconds, the angle for a chip shot is changing at a rate of 0.100 rad/sec, and after 2.5 seconds, the angle for a chip shot is changing at a rate of 0.179 rad/sec. For Part V (the through ball), it was discovered that after 2 seconds, the distance between the keeper and the offensive player is changing at a rate of -8.135 yards/second.

By completing this investigation, I analyzed a soccer game in a totally different way than what I was used to, and it opened my eyes to a whole new way of looking at a soccer game. To investigate the related rates in the breakaway, shot, through ball, and corner kick, I had to really think about the relationships between players, angles, the ball, and distance. Solving the related rate, and really analyzing why this related rate is important made me think about soccer in a mathematical way, and I quantified some of the instinctual and practiced knowledge a player has through evaluation. Dissecting a play down to the smallest detail, for example, the rate at which the angle for a shot is changing was new and intriguing. This related rate in particular really opened my eyes to the advice my coaches give me, and other players. When a player continues to dribble parallel to the goal, and more time passes, the decrease of theta increases, as seen in the comparison of $\frac{d\theta}{dt}$ evaluated after one second, and then evaluated after two seconds in Part II: The Shot. The angle for a shot to the far post is diminishing faster as the player continues to hold on to the ball. This was supported by the related rate since at 1 second, the angle for the shot is decreasing at a rate of -0.209 rad/sec, and at 2 seconds, the angle for the shot is decreasing at a rate of -0.238 rad/sec. Therefore, it makes sense that coaches urge players to shoot once they receive the ball, since for the best shot, theta should be greatest, and changing the slowest. Playing soccer is one of my favorite things to do, so investigating the math behind some of the most common offensive plays was really engaging.

This investigation did allow me to analyze a soccer game into more depth, and it made me more curious to question the mathematics behind the defensive side of the soccer game as well. Therefore, to improve upon the investigation above, another part would be added in which the related rates associated with defense would be analyzed. For example, it would really be interesting to used related rates to determine how defensive players time stepping and tackling a ball.

After investigating the related rates present in a soccer game, it made me curious as to

how related rates could be applied in other areas of my life, including my everyday lifestyle. For instance, what are the related rates in cooking? Or the related rates in digestion? This assessment also made me think about other areas that use related rates. For example, related rates could be used to analyze traveling. It would be interesting to investigate the relationships between planes, cars, and trains. Other related rates would be fascinating to investigate would be ones that have to do with the movement of objects in space. For example, the related rates between satellites, and something on earth. Overall, this assessment made me really analyze and inquire into the relationships in the game I love to play, and also other areas of interest.

Bibliography

Punis, Josip. "What are the official dimensions of a soccer field in the FIFA World Cup?" What are the official dimensions of a soccer field in the FIFA World Cup? - Quora. December 4, 2014. Accessed March 19, 2017. https://www.quora.com/What-are-the-official-dimensions-of-a-soccer-field-in-the-FIFA-World-Cup.

IB Mathematics SL: Exploration Feedback						
Title:	Related Rates in a Soccer Game		Date Set: March 24, 2017		Date Submitted: March 24, 2017	

A Communication	Score:	4	marks (out of 4 possible marks)

- Expressed ideas clearly
- Identified a clear aim for the exploration
- Focused on the aim and avoiding irrelevance
- Structured ideas in a logical manner
- Edited the exploration so that it is easy to follow
- Cited references where appropriate
- Included Introduction, rationale, description of aims, conclusion
- Graphs, tables and diagrams accompanied the work in the appropriate place and were not attached as appendices to the document.

4 The exploration is coherent, well organized, concise and complete.

Comments: Student clearly identifies aim- "should soccer players take the shot earlier depending on position of the player"- and expresses investigation thoroughly and logically. References are cited when using an internet source. Graphs, tables, and diagrams are used through the exploration in an appropriate manner.

B Mathematical presentation	Score:	3	marks (out of 3 possible marks)

- Used appropriate mathematical language and representation (notation, symbols, terminology)
- Defined key terms, where required
- Selected appropriate mathematical tools (including information and communication technology)
- Expressed results to an appropriate degree of accuracy
- Used multiple forms of mathematical representation, such as formulae, diagrams, tables, charts, graphs and models, where appropriate.
- Used mathematical language when communicating mathematical ideas, reasoning and findings.
- Chose and used appropriate ICT tools such as graphic display calculators, screenshots, graphing, spreadsheets, databases, drawing and word-processing software, as appropriate, to enhance mathematical communication.

3 The mathematical presentation is appropriate throughout.

Comments: Student appropriately uses mathematics with correct connotation and accuracy. Student models each situation on the soccer field and accurately documents personal findings in appropriate tables. Student has small mathematical error on page 3. Student should take d/dt of both sides. Student then corrects the mistake on the next line, and it ultimately does not affect finding. Student uses this same notation correctly on all other pages of exploration.

C Personal engagement	Score:	4	marks (out of 4 possible marks)

- Worked independently
- Asked questions, making conjectures and investigating mathematical ideas
- Read mathematics and researched areas of interest
- Looked for and creating mathematical models for real-world situations
- Considered historical and global perspectives
- Demonstrated that he or she has "made the exploration their own"
- Expressed ideas in an individual way
- The extent to which the student engaged with the exploration and made it their own.
- Thought independently and/or creatively
- Addressed personal interest
- Presented mathematical ideas in their own way.
- Evidence that the student made the work their own, not just simply a written report.
- Asked and answered questions: "I wonder if…", "What would happen if…" "Why does that happen…"
- The student's voice could be heard

4 There is abundant evidence of outstanding personal engagement.

Comments: Student's voice is clearly heard throughout exploration. Exploration is authentic because student does her own investigation with players on her soccer team in order to gather the data for her related rates. Her personal engagement is abundant when she states, "Therefore, the question arose…". Considers history of the sport in the development of coaching strategy.

D Reflection	Score:	3	marks (out of 3 possible marks)

- How the student reviewed, analyzed and evaluated the exploration.
- Discussed the implications of results
- Considered the significance of the exploration
- Looked at possible limitations and/or extensions
- Made links to different fields and/or areas of mathematics

3 There is substantial evidence of critical reflection.

Comments: She reflects throughout the paper. She considers the significance of her exploration, as she finds justification for her coach's advice. Her reflections and analysis lead her to new mathematical questions. She links her exploration to extending her project to analyze the defensive side of the ball, rather than just offensive. She begins to see related rates in the matrix of life- including cooking, travel, digestion, movement of objects in space- including satellites.

E Use of mathematics	Score:	6	marks (out of 6 possible marks)

- Demonstrated knowledge and understanding
- Produced work that is within the level of the course. It should not be solely based on mathematics listed in the prior learning.
- Applied mathematics in different contexts
- Applied problem-solving techniques
- Recognized and explained patterns, where appropriate
- This criterion assesses to what extent students used mathematics in the exploration.
- Sophistication in mathematics may include understanding and use of challenging mathematical concepts, looking at a problem from different perspectives and seeing underlying structures to link different areas of mathematics.
- Rigour involves clarity of logic and language when making mathematical arguments and calculations.
- Precise mathematics is error-free and uses an appropriate level of accuracy at all times.

6 Relevant mathematics commensurate with the level of the course is used. The mathematics explored is correct. Thorough knowledge and understanding are demonstrated.

Comments: Mathematics is beyond the level of the course. Solved multiple problems and perspective related to the offensive side of the soccer field. Demonstrated appropriate knowledge and understanding. Mathematically sound and accurate. Mastery of related rates concept. Logical and clearly mathematical arguments. Overall, truly impressive.

Total Marks Earned:	20	marks (out of 20 possible marks)